

Two barriers for minimal-time firing squads, and certified bounds for four and five states

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Abstract

A minimal-time solution of the firing squad synchronization problem fires every line of length n at time $2n - 2$. Every line agrees with the infinite half-line until the echo of its right end returns, and the rest of the line is a function of two anti-diagonals of the half-line. We prove two necessary conditions on the half-line of every minimal-time solution, whatever its number of states: its non-periodic part reaches the line $i = t/3$, and it is not periodic in any wedge at the left border; in particular it is never eventually regular. With a validated propositional encoding and LRAT-checked solver runs we obtain a certified proof that four states do not suffice, show that every five-state solution uses at least 15 distinct neighbourhoods on its half-line below the anti-diagonal $t + i = 78$, and confirm a conditional five-state result of Balzer (1967). A five-state rule that synchronizes all lines of length at most 14 shows that refutations along these lines need longer lines.

Keywords: firing squad synchronization problem, cellular automata, state complexity, minimal time, pumping lemma, proof certificates

1. Introduction

The firing squad synchronization problem, posed by Myhill in 1957 and popularised by Moore [1], asks for a finite automaton that, placed in every cell of a line of arbitrary length n and started with all cells quiescent except a *general* at one end, drives all cells into a firing state for the first time at the same instant. Goto [2] and Waksman [3] solved it in time $2n - 2$, which is optimal. The least number of states of a minimal-time solution is a benchmark question [4]: Waksman used 16 states, Balzer 8 [5], Gerken 7 [6] and Mazoyer 6 [7]. Balzer reported a computer search excluding four states [5]; Yunès [8] repeated it and observed that the lines of lengths $2, \dots, 9$ suffice, and Sanders [9] found Balzer's backtracking incomplete and confirmed the result with a corrected search. Whether five states suffice has been open since 1987 [4, 10]. Since the set of minimal-time solutions is not recursively enumerable [11], a negative answer would consist of necessary conditions that no five-state rule satisfies.

This note develops such conditions for any number of states and measures how far they reach. Until the echo of its right border returns, the line of length n behaves exactly like the infinite half-line, so a solution consists of one half-line diagram C_∞ and, for each n , a triangle R_n that is a function of two anti-diagonals of C_∞ (Section 2). Our results are: (i) a *pumping lemma* at the right end, which forces the non-periodic part of C_∞ to reach the line $i = t/3$ (Theorem 3.4) and explains a priori the speed-1/3 signal of the classical constructions; (ii) a *left-border lemma* with an explicit bound (Lemma 4.1), by which C_∞ is not periodic in any wedge at the left border (Theorem 4.2) and never eventually regular (Corollary 4.3); (iii) *certified bounds*: an LRAT certificate for the known fact that four states do not suffice (Theorem 5.1), whose earlier proofs rest on search programs, and for five states the bound $c_{78} \geq 15$ (Proposition 5.4) and Balzer's conditional result

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(Proposition 5.5); (iv) a five-state rule for all lengths up to 14 (Proposition 5.2) and a left-border refutation of the half-line of rules for the lengths up to 12 and 13 (Corollary 5.3). Items (i) and (ii) are theorems, (iii) are unsatisfiability claims with independently checked LRAT proofs, (iv) are finite simulations.

Related work. Pigeonhole arguments on repeated pairs of states exclude minimal-time solutions of variants of the problem [12] and with one-way channels [11, Prop. 2]; our pumping lemma applies this mechanism to the original problem, where it constrains solutions instead of excluding them. The periodicity behind the front used in Theorem 3.4 also underlies the work of Mazoyer and Terrier on signals [13]. Settle [14] excludes four- and five-state solutions under additional constraints.

2. Lines and the half-line

Let S be a finite set of *states* containing the quiescent state L , the general G and the firing state F ; put $W := S \setminus \{F\}$ and let $*$ $\notin S$ be the border. A *rule* is a map $\delta : (W \cup \{*\}) \times W \times (W \cup \{*\}) \rightarrow S$ with

$$\delta(L, L, L) = \delta(L, L, *) = L. \quad (1)$$

The *line of length n* is $C_n : \mathbb{N} \times \{1, \dots, n\} \rightarrow S$ with $C_n(0, 1) = G$, $C_n(0, i) = L$ for $i \geq 2$ and $C_n(t+1, i) = \delta(C_n(t, i-1), C_n(t, i), C_n(t, i+1))$, where $C_n(t, 0) = C_n(t, n+1) = *$, considered up to the first time at which some cell is F . The rule *synchronizes* it at time T if all cells are F at time T and none before. A *minimal-time solution* synchronizes the line of length n at time $2n-2$ for every $n \geq 2$; it has k states if $|S| = k$. The *half-line* $C_\infty : \mathbb{N} \times \{1, 2, \dots\} \rightarrow S$ is defined in the same way with a border on the left only.

Lemma 2.1. *For every rule, $C_n(t, i) = C_\infty(t, i) = L$ whenever $t < i-1$ (and $i \leq n$ for C_n).*

Proof. Induction on t ; for $t = 0$ this is the initial condition. For $t \geq 1$ the predecessors $(t-1, i')$, $i' \in \{i-1, i, i+1\}$, satisfy $t-1 < i'-1$, so they are L or the right border, and (1) gives L . $\square$

Lemma 2.2 (agreement). *For every rule and $n \geq 2$: $C_n(t, i) = C_\infty(t, i)$ whenever $1 \leq i \leq n$ and $t+i \leq 2n-2$.*

Proof. Induction on t , the case $t = 0$ being clear. If $i = n$, then $t \leq n-2 < i-1$ and both sides are L by Lemma 2.1. If $i < n$, the predecessors $(t-1, j)$, $j \in \{i-1, i, i+1\}$, satisfy $(t-1) + j \leq 2n-2$ and $j \leq n$, so they agree by induction (for $j = 0$ both are the border). $\square$

The border is felt only in the *reflected triangle* $R_n := \{(t, i) : 1 \leq i \leq n, t+i \geq 2n-1, t \leq 2n-2\}$.

Lemma 2.3 (return chain). *If δ synchronizes the lines of lengths n and $n+1$ at times $2n-2$ and $2n$, then $C_n(2n-1-i, i) \neq C_\infty(2n-1-i, i)$ for $1 \leq i \leq n$, and the front $f_i := C_\infty(i-1, i)$ satisfies $f_{n-1} \neq L$.*

Proof. For $i = 1$, $C_n(2n-2, 1) = F$, whereas $C_\infty(2n-2, 1) = C_{n+1}(2n-2, 1) \neq F$ by Lemma 2.2 and since $2n-2 < 2n$. If the claim holds for $i < n$, then the left and middle predecessors of $(2n-1-i, i)$ have coordinate sums $2n-3$ and $2n-2$ and agree in C_n and C_∞ (for $i = 1$ the left one is the border), so the right ones differ: this is the claim for $i+1$. For $i = n$ it reads $\delta(f_{n-1}, L, *) \neq \delta(f_{n-1}, L, L)$ (by Lemmas 2.1 and 2.2), which is impossible if $f_{n-1} = L$ by (1). $\square$

By Lemmas 2.1 and 2.2, a rule that synchronizes every line cannot synchronize the line of length n before time $2n-2$ [3]. In a minimal-time solution every cell i leaves L exactly at time $i-1$ (Lemma 2.3), and since $C_\infty(t, i) = C_n(t, i)$ for n large and the line of length n does not fire before time $2n-2$,

$$C_\infty(t, i) \neq F \quad \text{for all } t \geq 0, i \geq 1. \quad (2)$$

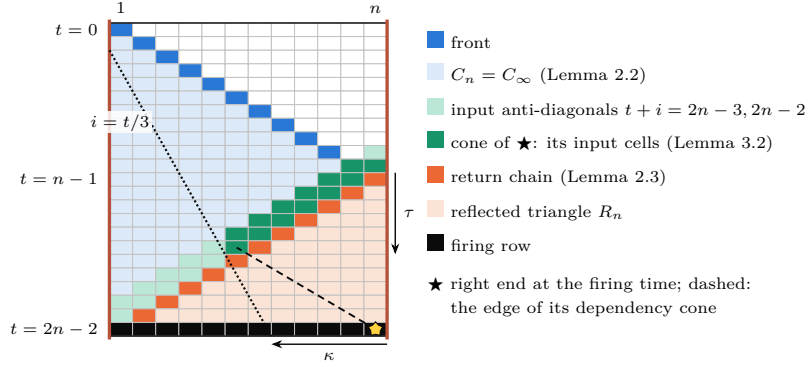


Figure 1: The line of length $n = 12$ (time downwards; colours show regions, not states). R_n is computed from the two input anti-diagonals (Lemma 3.1); the firing of the right end ($\star$) depends only on the dark input cells (Lemma 3.2), which end where the line $i = t/3$ crosses the input anti-diagonals (Theorem 3.4).

3. The right end: a speed-1/3 barrier

Fix $n \geq 2$ and use the relative coordinates $\tau := t - (n - 1)$, $\kappa := n - i$, in which $R_n = \{(\tau, \kappa) : 0 \leq \kappa \leq \tau \leq n - 1\}$, with the right border at $\kappa = -1$ and the left border at $\kappa = n$. The predecessors of a cell of R_n lie in R_n , on a border, or on one of the anti-diagonals $t + i = 2n - 3$, $t + i = 2n - 2$, where $C_n = C_\infty$. The *input word* of R_n is

$$J_n(\kappa) := (C_\infty(n - 3 + \kappa, n - \kappa), C_\infty(n - 2 + \kappa, n - \kappa)), \quad 0 \leq \kappa \leq n - 1,$$

with *lower* and *upper input cells* on $t + i = 2n - 3$ and $2n - 2$ (for $n = 2$, $\kappa = 0$ the lower cell lies at time -1 , is a predecessor of no cell of R_n , and is set to L); see Figure 1.

Lemma 3.1 (determinacy). *There is a partial map Ψ , depending on δ only, such that $C_n(n - 1 + \tau, n - \kappa) = \Psi(J_n, n; \tau, \kappa)$ for all $n \geq 2$ and $(\tau, \kappa) \in R_n$, and $\Psi(J, N; \tau, \kappa)$ depends only on the input cells and borders in the backward dependency cone of (τ, κ) .*

Proof. Define $\Psi(J, N; \tau, \kappa)$ for $0 \leq \kappa \leq \tau \leq N - 1$ by induction on τ as $\delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1})$, where $x_{\kappa'}$ is $*$ if $\kappa' \in \{-1, N\}$, the lower or upper component of $J(\kappa')$ if $\tau - 1 = \kappa' - 2$ or $\tau - 1 = \kappa' - 1$, and $\Psi(J, N; \tau - 1, \kappa')$ otherwise (then $0 \leq \kappa' \leq \tau - 1$). When $N = n$, these are exactly the left, middle and right predecessors of the cell $(n - 1 + \tau, n - \kappa)$, with $C_n = C_\infty$ on the input cells, so induction on τ gives the first claim; unfolding the definition gives the second. $\square$

For $(\tau, \kappa) \in R_n$ let $\text{cone}_n(\tau, \kappa)$ be the set of input cells reached from (τ, κ) by repeatedly passing to predecessors, stopping at input cells and borders.

Lemma 3.2. *For $n \geq 4$, $\text{cone}_n(n - 1, 0)$ avoids the left border and consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \leq \kappa' \leq \lfloor n/2 \rfloor$ and, if n is odd, the lower one at $\kappa' = (n + 1)/2$.*

Proof. A step changes τ by -1 and κ by at most one, and input cells satisfy $\tau' \in \{\kappa' - 1, \kappa' - 2\}$; a path from $(n - 1, 0)$ reaching an input cell at κ' after s steps has $\kappa' \leq s$ and $n - 1 - s \in \{\kappa' - 1, \kappa' - 2\}$, so $2\kappa' \leq n + 1$, and $2\kappa' \leq n$ for upper cells. Conversely, the upper cell at $\kappa' \leq n/2$ is reached by κ' left moves followed by vertical moves, and the lower cell at $1 \leq \kappa' \leq (n + 1)/2$ by $\kappa' - 1$ left moves, vertical moves down to $(\kappa' - 1, \kappa' - 1)$ and one more left move, through cells of R_n . The lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n , and $\kappa = n$ would require n left moves, i.e. a negative time. $\square$

Lemma 3.3 (pumping). *Let δ be a minimal-time solution and $4 \leq n < n'$. Then J_n and $J_{n'}$ differ on $\text{cone}_n(n - 1, 0)$.*

Proof. Otherwise, by Lemmas 3.1 and 3.2, the relative cell $(n-1, 0)$ of $R_{n'}$ (it lies in $R_{n'}$, and its cone has the same positions, far from the left border $\kappa = n'$) has the value of the relative cell $(n-1, 0)$ of R_n , namely $C_n(2n-2, n) = F$. Thus $C_{n'}(n' + n - 2, n') = F$ although $n' + n - 2 < 2n' - 2$. $\square$

The lemma constrains C_∞ alone. In the frame of the front, $\Phi_t(j) := C_\infty(t, t+1-j)$ ($0 \leq j \leq t$) is the cell at *depth* j behind the front, and the recurrence becomes

$$\Phi_{t+1}(j) = \delta(\Phi_t(j), \Phi_t(j-1), \Phi_t(j-2)), \quad \Phi_t(-1) = \Phi_t(-2) = L, \quad (3)$$

so information only travels to larger depths. In these coordinates $J_n(\kappa) = (\Phi_{n-3+\kappa}(2\kappa-2), \Phi_{n-2+\kappa}(2\kappa-1))$.

Theorem 3.4 (speed-1/3 barrier). *Let δ be a minimal-time solution. Every depth-row $t \mapsto \Phi_t(j)$ is eventually periodic; let $\theta(j)$ be the time from which it is periodic. For every $n \geq 4$ there is a depth $j \leq n$ with $\theta(j) > n + \frac{j}{2} - \frac{5}{2}$. Hence $\limsup_{j \rightarrow \infty} \theta(j)/j \geq 3/2$: along infinitely many depth-rows the transient part extends, up to an arbitrarily small angle, to the line $i = t/3$.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), L, L)$ and is eventually periodic; by (3), row j is a finite-state system driven by the eventually periodic rows $j-1, j-2$, hence eventually periodic, with some period P_j (cf. [13, §4]). Fix $n \geq 4$ and a common multiple P of $P_0, \dots, P_{n+1}$. At $\kappa = 0$ the input cells of R_n and R_{n+P} are quiescent (Lemma 2.1). If $n-3+\kappa \geq \theta(2\kappa-2)$ and $n-2+\kappa \geq \theta(2\kappa-1)$ for all cone positions $\kappa \geq 1$, then $J_{n+P} = J_n$ on the cone, contradicting Lemma 3.3. Since cone positions satisfy $\kappa \leq \lceil n/2 \rceil$, the violated inequality concerns a depth $j \in \{2\kappa-2, 2\kappa-1\}$ with $j \leq n$ and $\theta(j) > n-3+\kappa \geq n + \frac{j}{2} - \frac{5}{2}$. If $\limsup \theta(j)/j < 3/2$, then $\theta(j) \leq \alpha j + \beta$ for all j with some $\frac{1}{2} \leq \alpha < \frac{3}{2}$ and β , and for n large and all $j \leq n$ we get $(\alpha - \frac{1}{2})j \leq (\alpha - \frac{1}{2})n < n - \beta - \frac{5}{2}$, i.e. $\theta(j) \leq n + \frac{j}{2} - \frac{5}{2}$, a contradiction. Finally, the cell of depth j at time $\theta(j)$ is at position $\theta(j) + 1 - j$, which is at least $\theta(j)/3 + 1$ when $\theta(j) \geq 3j/2$. $\square$

The speed-1/3 signal of Waksman's and Balzer's constructions runs along the line $i = t/3$; in Mazoyer's solution the periodic zone behind the front ends at speed 1/2 (for $1 \leq j < 700$, $\theta(j) \in \{2j+1, 2j+2\}$ with period 3).

4. The left end: a second barrier

Near the left border the reflected triangle is produced by a band of anti-diagonals behind the reflected wave, which acts like a finite automaton reading the half-line and must fire each cell exactly when it arrives there. For $c \geq -1$ and $\max(1, c) \leq i \leq n$ put $v_c(i) := C_n(2n-2+c-i, i)$, the cell of the *chain* $t+i = 2n-2+c$ at position i ; chains -1 and 0 are the input anti-diagonals and chain 1 is the return chain. The predecessors of $(2n-2+c-i, i)$ lie on the chains $c-2, c-1, c$, so

$$v_c(i) = \delta(v_{c-2}(i-1), v_{c-1}(i), v_c(i+1)) \quad (c \geq 1), \quad (4)$$

with $v_{c-2}(0)$ and $v_c(n+1)$ standing for the border. If the line of length n is synchronized at time $2n-2$, then $v_i(i) = F$ and $v_c(i) \neq F$ for $-1 \leq c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$ this becomes

$$u_c(y) = \delta(u_{c-2}(y), u_{c-1}(y + \varepsilon_c), u_c(y+1)), \quad \varepsilon_c = 1 \text{ for even } c, \varepsilon_c = 0 \text{ for odd } c, \quad (5)$$

with $u_0(y) = C_\infty(2n-2-y, y)$ and $u_{-1}(y) = C_\infty(2n-2-y, y-1)$; here $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n-2 + \lfloor c/2 \rfloor - y$, which is F if $\lfloor c/2 \rfloor = y$ and not F if $\lfloor c/2 \rfloor < y$. Each slice $(u_1(y), u_2(y), \dots)$ is thus computed from the slice $y+1$ and the two input cells $u_{-1}(y), u_0(y)$ (Figure 2).

Lemma 4.1 (left border). *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $s \geq 1$, $L \geq 1$, $s+L \leq Y \leq n-s$, and suppose that the words $y \mapsto C_\infty(2n-2-y, y)$ ($s \leq y \leq Y$) and $y \mapsto C_\infty(2n-3-y, y)$ ($s \leq y \leq Y-1$) are L -periodic. Then $Y-s \leq (k-1)^{2s}L$.*

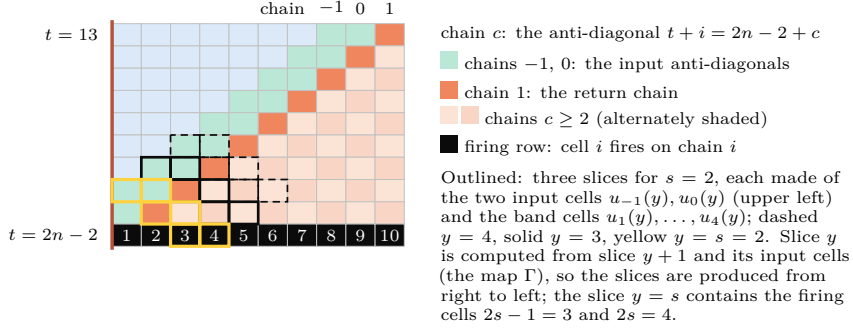


Figure 2: The lower left corner of the line of length $n = 12$ (time downwards; colours show regions, not states), with the chains and three of the slices used in the proof of Lemma 4.1.

Proof. For $s \leq y \leq Y$ let $\mathbf{u}(y) := (u_1(y), \dots, u_{2s}(y))$; these are cells $y + \lfloor c/2 \rfloor \leq n$ with $\lceil c/2 \rceil \leq s \leq y$, so they are defined, and none is F when $y > s$. By (5), $u_c(y)$, $1 \leq c \leq 2s$, is computed from components of index $< c$ of $\mathbf{u}(y)$, of index $\leq c$ of $\mathbf{u}(y+1)$, and from $u_{-1}(y), u_0(y)$; hence $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y))$ for $s+1 \leq y \leq Y-1$, with Γ depending on δ only. Unfolding (5) for even c , $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y) = \Gamma'(\mathbf{u}(y+1), u_0(y))$ for $s \leq y \leq Y-1$. By hypothesis, u_0 is L -periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose $Y - s > (k-1)^{2s}L$. The pairs $(\mathbf{u}(y), y \bmod L)$, $s+1 \leq y \leq Y$, take at most $(k-1)^{2s}L$ values, so there are $s+1 \leq y_1 < y_2 \leq Y$ with $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ and $L \mid h := y_2 - y_1$. By downward induction, $\mathbf{u}(y) = \mathbf{u}(y+h)$ for $s+1 \leq y \leq y_1$: if it holds for $y+1 \leq y_1$, then $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y)) = \Gamma(\mathbf{u}(y+1+h), u_{-1}(y+h), u_0(y+h)) = \mathbf{u}(y+h)$, as $y+h < y_2 \leq Y$. Since $s+h \leq Y-1$ and $u_0(s+h) = u_0(s)$,

$$u_{2s}(s+h) = \Gamma'(\mathbf{u}(s+1+h), u_0(s+h)) = \Gamma'(\mathbf{u}(s+1), u_0(s)) = u_{2s}(s) = F,$$

although $\lceil 2s/2 \rceil = s < s+h$, so that $u_{2s}(s+h)$, the cell $2s+h$ at time $2n-2-h$, is not F: a contradiction. $\square$

Theorem 4.2 (left-border barrier). *Let δ be a minimal-time solution. There are no $s \geq 1$, $\alpha > 0$, t_0 and $p, q \geq 1$ such that $C_\infty(t+p, i) = C_\infty(t, i)$ and $C_\infty(t, i+q) = C_\infty(t, i)$ whenever the cells involved lie in the wedge $\Omega := \{(t, i) : s \leq i \leq \alpha t, t \geq t_0\}$.*

Proof. Suppose otherwise and put $L := pq$. If (t, i) and $(t-L, i+L)$ lie in Ω , so do the cells $(t-L, i+jq)$, $0 \leq j \leq p$, and $(t-L+jp, i)$, $0 \leq j \leq q$, hence $C_\infty(t-L, i+L) = C_\infty(t-L, i) = C_\infty(t, i)$. Let $Y_n := \lfloor \alpha(2n-3)/(1+\alpha) \rfloor$. For $s \leq y \leq Y_n$ the cells $(2n-2-y, y)$ and $(2n-3-y, y)$ satisfy $y \leq \alpha(2n-3-y)$, so they lie in Ω once $2n-3-Y_n \geq t_0$, i.e. for n large; since a step of L positions along an anti-diagonal is the step from (t, i) to $(t-L, i+L)$, the words of Lemma 4.1 are L -periodic on $[s, Y_n]$ and $[s, Y_n-1]$. As $Y_n \geq \frac{2\alpha}{1+\alpha}n - 4$, for n large $Y := \min(Y_n, n-s)$ satisfies $Y \geq s+L$ and $Y-s > (k-1)^{2s}L$, contradicting Lemma 4.1. $\square$

The half-line of a rule is *eventually regular* if there are t_1 , $P \geq 1$ and, for each residue $r \in \{0, \dots, P-1\}$, words $w_0, \dots, w_m$, nonempty words $z_1, \dots, z_m$ over W and integers $a_j, D_j \geq 0$ such that for all $\nu \geq 0$ the configuration $C_\infty(t, 1) \cdots C_\infty(t, t+1)$ at time $t = t_1 + r + \nu P$ is $w_0 z_1^{a_1 + \nu D_1} w_1 \cdots z_m^{a_m + \nu D_m} w_m$: finitely many periodic regions separated by signals of rational speeds.

Corollary 4.3. *The half-line of a minimal-time solution is not eventually regular.*

Proof. Suppose it is. For each residue r some D_j is positive, because the configuration grows by P cells per period; let $j(r)$ be the least such index. The blocks and junctions left of $z_{j(r)}$ have constant lengths, so the block $z_{j(r)}^{a_{j(r)} + \nu D_{j(r)}}$ starts at a fixed cell s_r and its length grows at least like βt , for some $\beta > 0$ and t large. Inside it the state of a cell depends only on its distance to s_r modulo $|z_{j(r)}|$, and is the same at the times t and $t+P$. With $s := \max_r s_r$, $q := \prod_r |z_{j(r)}|$, $\alpha := \beta/2$ and t_0 large, every cell of $\{(t, i) : s \leq i \leq \alpha t, t \geq t_0\}$

lies in the block of index $j(r)$ of its residue class, so C_∞ is periodic with periods P in time and q in space on this wedge, contradicting Theorem 4.2. $\square$

Remark 4.4. The proof of Lemma 4.1 uses only (4) for $c \geq 1$, the firing condition and the periodicity of the chains $c = -1, 0$. Applied to the mirror image of the line, it shows: if δ synchronizes the line of length n at time $2n - 2$, $s + L \leq Y \leq n - s$ and $Y - s > (k - 1)^{2s}L$, then the words $y \mapsto C_n(2n - 2 - y, n + 1 - y)$ ($s \leq y \leq Y$) and $y \mapsto C_n(2n - 3 - y, n + 1 - y)$ ($s \leq y \leq Y - 1$) are not both L -periodic; for $2y < n$ these cells lie in R_n . In the classical constructions the structure required by Theorem 4.2 is the family of signals of speeds $1/(2^j - 1)$ accumulating at the border [3, 5], or the recursive division points [7].

5. Certified bounds for four and five states

The CNF formula $\text{MT}_k(\mathcal{N}; M, N')$ has one-hot variables for the entries of a k -state rule table, for the cells of C_∞ with $t + i \leq M$ and for the cells of R_n , $n \in \mathcal{N}$. Its clauses say that these cells follow the rule, that no half-line cell fires (2), and that the line of length $n \in \mathcal{N}$ fires at time $2n - 2$ and not before; they add the return chains (Lemma 2.3), the constraints $\text{PUMP}(n, n')$ of Lemma 3.3 for $4 \leq n < n' \leq N'$, clauses implied by Lemma 3.1 and a symmetry breaking on the auxiliary states. All families follow from the definition of minimality and the lemmas above, so the formula is satisfiable if a k -state minimal-time solution exists. Mazoyer’s solution, extracted from Duprat’s Coq proof of its correctness [15], satisfies $\text{MT}_6(\{3, \dots, 12\}; 78, 40)$, and all models were re-simulated by an independent checker. Every unsatisfiability claim below comes with an LRAT proof [16] from CaDiCaL [17], or from Kissat [18] via `drat-trim`, checked by `lrat-check` [19].

Theorem 5.1. *No four-state rule satisfying (1) synchronizes all lines of lengths $2, \dots, 9$ in minimal time; in particular no four-state minimal-time solution exists.*

Proof. Such a rule yields a model of $\text{MT}_4(\{2, \dots, 9\}; 16)$: a half-line cell with $t + i \leq 16$ is quiescent if $i \geq 10$ (Lemma 2.1) and equals a cell of the line of length 9 at a time $t < 16$ if $i \leq 9$ (Lemma 2.2), so it does not fire, and the return chains hold by Lemma 2.3 for $2 \leq n \leq 8$. CaDiCaL refutes the formula (753 variables, 17,605 clauses) in 51 s with an LRAT proof of 177 MB, which `lrat-check` verifies in 5.7 s. $\square$

The bound 9 is sharp: an exhaustive enumeration finds 27 partial rules, counted on the transitions they use, for the lengths $2, \dots, 8$, in agreement with [8, 9].

The five-state frontier. A five-state table has 94 free entries with five values (four states: 43 with four values). Every instance $\text{MT}_5(\mathcal{N}; M, N')$ that our solvers decided was satisfiable, and each rule returned failed at the first length not in $\mathcal{N}$.

Proposition 5.2. *The five-state rule δ_{14} of Table 1 synchronizes the line of length n at time $2n - 2$ for every $2 \leq n \leq 14$; on the line of length 15 the cells 13, 14, 15 fire at time 22. Its half-line uses 16 neighbourhoods besides (L, L, L) on the cells with $t + i \leq 10^4$, none of them firing, and satisfies $\text{PUMP}(n, n')$ for $4 \leq n < n' \leq 800$; for $n \leq 5000$, $s \leq 3$ and $L \leq 24$, Lemma 4.1 does not apply to its input anti-diagonals.*

Proof. Direct simulation, by programs independent of the encoder. $\square$

For $3 \leq t < 1500$ the half-line of δ_{14} is in states L, G at cell 1, L, A at cell 2 and L, B elsewhere, where it follows the elementary cellular automaton rule 22 [20]: it is chaotic. Four further rules, on other half-lines, also synchronize all lengths up to 14 and fail at 15. Hence our necessary conditions restricted to the lengths up to 14 are satisfiable, and refutations along these lines need lines of length at least 15.

The left-border lemma does refute a more regular architecture. The rules δ_{12} , δ_{13} of the repository synchronize every line of length at most 12, resp. 13; their half-lines coincide for $t + i \leq 78$, use there 22 neighbourhoods besides (L, L, L), and consist of a front and a wake of G’s and, behind a boundary of speed $1/3$ (where Theorem 3.4 allows the transient part to end), a region $\text{A L A L} \dots$ periodic in time and space.

x	y	$*$	L	G	A	B	x	y	$*$	L	G	A	B
$*$	L	–	F	F	L	G	$*$	A	–	G	G	L	G
L	L	L	L	G	G	B	L	A	A	B	G	B	A
G	L	G	A	B	A	B	G	A	B	A	G	L	A
A	L	G	B	G	G	G	A	A	B	L	A	G	A
B	L	A	B	A	A	L	B	A	L	A	A	G	A
$*$	G	–	L	F	L	L	$*$	B	–	G	G	G	G
L	G	F	A	F	B	A	L	B	G	B	G	A	L
G	G	B	B	F	G	F	G	B	F	B	A	B	B
A	G	B	A	G	G	A	A	B	G	B	B	G	B
B	G	B	A	G	A	L	B	B	A	L	A	A	L

Table 1: The rule δ_{14} of Proposition 5.2: the entry in row (x, y) and column z is $\delta(x, y, z)$; “–” marks the unused neighbourhoods $(*, y, *)$. Machine-readable copy: `results/delta14.txt` in the repository.

Corollary 5.3. *No rule that agrees with δ_{12} on these 22 neighbourhoods synchronizes the line of length 518 in minimal time.*

Proof. With these neighbourhoods and (L, L, L) only, no new neighbourhood occurs on the cells with $t + i \leq 1034$, so every such rule has the same diagram C_∞ there. On $t + i = 1034$ the cells $2, \dots, 259$ are in state A, and on $t + i = 1033$ the cells $2, \dots, 258$ are in state L. Lemma 4.1 with $k = 5$, $n = 518$, $s = 2$, $L = 1$, $Y = 259$ would give $257 \leq 4^4 = 256$. $\square$

Half-lines of low complexity. The germ below M of a rule is its restriction to the neighbourhoods of the cells (t, i) , $t \geq 1$, $t + i \leq M$, of its half-line, and its complexity $c_M(\delta)$ is the number of these neighbourhoods other than (L, L, L): $c_{78} = 57$ for Mazoyer’s rule, 22 for δ_{12} and 16 for δ_{14} .

Proposition 5.4. *Every five-state minimal-time solution δ satisfies $c_{78}(\delta) \geq 15$.*

Proof (computer-assisted). Suppose $c_{78}(\delta) \leq 14$. After exchanging A and B if necessary, A occurs before B (if at all) when the cells are ordered by anti-diagonal and, within an anti-diagonal, by time. By (2) and Lemmas 2.3 and 3.3, the germ of δ below 78 produces no firing cell, never produces L at the front, and satisfies PUMP(n, n') for $4 \leq n < n' \leq 40$. A depth-first search that chooses the value of a neighbourhood when it first occurs enumerates all such germs: $1.0 \cdot 10^8$ nodes, 244 germs, and an independent re-implementation finds the same 244. For 230 of them, $\text{MT}_5(\{2, \dots, 10\}; 78, 39)$ with the germ is unsatisfiable (one formula with a selector variable, refuted in 4s with an LRAT proof of 11 MB). Each of the other 14 determines C_∞ up to the anti-diagonal 3300, and on these diagrams Lemma 4.1 applies with $s = 1$: there are $n \leq 163$ and $L \leq 5$ such that the input anti-diagonals of the line of length n are L -periodic on $[1, Y]$ with $Y - 1 > 16L$ (e.g. $n = 28$, $L = 1$, $Y = 19$), as an independent simulation confirms. Hence no rule containing one of the 244 germs, in particular δ , is a minimal-time solution. $\square$

Of the 9,787 germs with $c_{78} \leq 16$, the same pipeline (completed by a test for the lengths $2, \dots, 13$ and by Corollary 4.3 applied through a finite certificate of eventual regularity, see the extended version) leaves 16, among them that of δ_{14} , irregular as far as we can tell.

Balzer’s conditions. Balzer [5, p. 37] reported that no five-state minimal-time solution satisfies the four conditions (B1) $\delta(x, G, y) = G$ unless $x, y \in \{G, *\}$; (B2) $\delta(I(z), I(y), I(x)) = I(\delta(x, y, z))$ for an involution I of the states with $I(*) = *$ (an *image solution*); (B3) $\delta(G, G, G) = \delta(*, G, G) = \delta(G, G, *) = F$ and no other neighbourhood gives F; (B4) $\delta(G, v, G) = G$ for all $v \in W \setminus \{G\}$. His eight-state solution satisfies them, Mazoyer’s only (B3) [21]. In the *weak reading* [8], (B1) is required only for working states x, y and (B3) without its last clause. Taking $I(F) = F$, (B3) forces $I(G) = G$, so I is one of the four involutions of $\{L, A, B\}$, and (B1)–(B4) become unit and binary clauses. In the weak reading, no rule with $I = (AB)$ synchronizes the lengths $2, \dots, 10$, none with $I = (LA)$ or (LB) the lengths $2, \dots, 9$, and none with I the identity the lengths

$2, \dots, 12$; the largest LRAT proof, obtained from a Kissat proof by `drat-trim`, has 3.1 GB and is verified in 66 s. Rules with I the identity satisfying (B1)–(B4) in both readings synchronize every line of length at most 11.

Proposition 5.5. *No five-state rule satisfying (B1)–(B4) in the weak reading, with $I(F) = F$, synchronizes all lines of lengths $2, \dots, 12$ in minimal time; a fortiori none satisfying Balzer’s (strong) form does. The lengths up to 11 do not suffice to prove this.*

With six states, rules satisfying (B1)–(B4) in the strong reading exist for all lengths up to 13 for every type of involution, so Balzer’s conditions do not lead to a refutation of this kind there.

6. Concluding remarks

A minimal-time solution needs non-periodic structure along the line $i = t/3$ and in every wedge at the left border; the known constructions provide it with hierarchies of slow signals or division points, and the five-state half-lines that survive our tests are chaotic. Natural questions are: (1) Does a five-state minimal-time solution exist? A refutation along the lines of Theorem 5.1 needs lines of length at least 15. (2) Is there a necessary condition like Lemma 3.3 on the reflected triangles themselves, beyond Remark 4.4? (3) Can the bound $(k-1)^{2s}L$ of Lemma 4.1 be made polynomial in s , or extended to anti-diagonals that are periodic up to a bounded number of defects? (4) Do Balzer’s conditions exclude six states?

Declarations

CRedit. All authors: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Writing – review & editing. *Competing interests*. None. *Funding*. None. *Data availability*. Programs, formula generators, rule tables, logs, the SHA-256 digests of the formulas and certificates, the commands that regenerate the certificates, and an extended version of this note are at <https://github.com/giamkiakeng/proj1> (directory `fssp5`). *Use of generative AI*. The authors used Claude (Anthropic) to develop and check the mathematical arguments, to write and run the programs, to carry out the experiments and to draft and proofread the text; they reviewed and edited the content and take full responsibility for it.

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